THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. An egg is fertilised on the 14th day of the menstrual cycle. The next menstruation will be: 1

(a) on the 15th day **(b)** on the 21st day **(c)** on the 28th day **(d)** absent

2. Which of these processes removes nitrogen gas from the air for plants? 1

(a) nitrogen fixation **(b)** respiration

(c) transpiration **(d)** burning

3. The predictions of Newton's law of gravitation matched the observed motion of planets. This: 1

(a) proves it forever **(b)** refutes it

(c) has no effect **(d)** supports the law

4. Which technique separates a mixture of iodine and sand? 1

(a) decantation **(b)** magnet **(c)** sublimation **(d)** filtration

5. The formula of ammonium phosphate is: 1

(a) NH_4PO_4 **(b)** $(NH_4)_2PO_4$ **(c)** $(NH_4)_3PO_4$ **(d)** $NH_4(PO_4)_3$

6. An element with atomic number 14 has valency: 1

(a) 8 **(b)** 4 **(c)** 2 **(d)** 3

7. A car travels half the distance at 40 km/h and the other half at 60 km/h. The average speed is: 1

(a) 48 km/h **(b)** 45 km/h **(c)** 50 km/h **(d)** 52 km/h

8. A person weighs 600 N on the Earth (g = 10). On a planet where g is one-third of that on Earth, the mass of the person is: 1

(a) 20 kg **(b)** 60 kg **(c)** 600 kg **(d)** 180 kg

9. A body of mass 4 kg is lifted through 3 m and then carried horizontally 5 m. The total work done against gravity is (g = 10): 1

(a) 0 **(b)** 200 J **(c)** 320 J **(d)** 120 J

10. A wave of wavelength 2 m and speed 340 m/s has a time period of: 1

(a) 0.00588 s **(b)** 170 s **(c)** 680 s **(d)** 0.0059 Hz

11. An animal cell placed in pure water will: 1

(a) swell and may burst **(b)** remain same

(c) shrink **(d)** divide

12. A student sees dead cells with thick lignified walls in a section. The tissue is: 1

(a) collenchyma **(b)** sclerenchyma **(c)** phloem **(d)** parenchyma

13. An organism is multicellular, has a cell wall of chitin and feeds on dead matter. It is a: 1

(a) animal **(b)** fungus **(c)** plant **(d)** bacterium

14. A gardener wants a rose plant with flowers of two colours. The method is: 1

(a) grafting **(b)** binary fission **(c)** seed sowing **(d)** spore formation

15. Which of the following reduces your carbon footprint the most? 1

(a) using public transport and saving energy **(b)** burning waste

(c) using air conditioners all day **(d)** using plastic

16. 72 km/h in m/s is: 1

(a) 7.2 **(b)** 72 **(c)** 200 **(d)** 20

17. In fractional distillation the component with the lowest boiling point: 1

(a) stays in the flask **(b)** is not collected

(c) is collected first **(d)** is collected last

18. Which of the following substances will conduct electricity in the solid state? 1

(a) sodium chloride **(b)** sugar **(c)** copper **(d)** sulphur

19. Boron has isotopes 10 (20 per cent) and 11 (80 per cent). The average atomic mass is: 1

(a) 10.5 u **(b)** 11 u **(c)** 10.8 u **(d)** 10.2 u

20. A body in circular motion of radius 2 m makes 1 revolution in 4 s. Speed is (pi = 3.14): 1

(a) 1.57 m/s **(b)** 3.14 m/s **(c)** 6.28 m/s **(d)** 12.56 m/s

21. A block on a rough floor can resist a push of up to 12 N by static friction. If it is pushed by a force of 8 N and does not move, the friction force is: 1

(a) 12 N **(b)** 0 **(c)** 8 N **(d)** 4 N

22. A 60 W lamp is used for 5 hours daily for 30 days. At 5 rupees per unit the cost of the energy is: 1

(a) 450 rupees **(b)** 225 rupees **(c)** 9 rupees **(d)** 45 rupees

23. If the amplitude of a sound wave is doubled, the loudness: 1

(a) becomes zero **(b)** decreases **(c)** stays same **(d)** increases

24. A cell with 8 chromosomes divides by meiosis. Each daughter cell has: 1

(a) 8 **(b)** 4 **(c)** 2 **(d)** 16

25. A tissue of long cells with cilia on the surface is found in the: 1

(a) trachea **(b)** skin **(c)** kidney **(d)** stomach

26. An animal that lays eggs, has feathers and a beak is a: 1

(a) mammal **(b)** amphibian **(c)** reptile **(d)** bird

27. Papaya and cucumber flowers are: 1

(a) bisexual **(b)** unisexual **(c)** asexual **(d)** sterile

28. If the Earth received 1000 W per square metre and reflected 30 per cent, the radiation absorbed is: 1

(a) 300 W per square metre **(b)** 1030 W per square metre

(c) 700 W per square metre **(d)** 970 W per square metre

29. A car travels 90 km in 1.5 hours. Its speed in m/s is: 1

(a) 90 m/s **(b)** 60 m/s **(c)** 6 m/s **(d)** 16.7 m/s

30. The volume of ethanol in 250 mL of a 20 per cent (v/v) solution is: 1

(a) 5 mL **(b)** 50 mL **(c)** 200 mL **(d)** 20 mL

31. A compound has 40 per cent sulphur and 60 per cent oxygen by mass. 20 g of sulphur combines with how much oxygen? 1

(a) 12 g **(b)** 60 g **(c)** 30 g **(d)** 40 g

32. The atom with the configuration 2, 8, 8, 2 is: 1

(a) calcium **(b)** potassium **(c)** magnesium **(d)** argon

33. A car brakes from 20 m/s to rest with retardation 5 m/s squared. The distance covered is: 1

(a) 4 m **(b)** 10 m **(c)** 40 m **(d)** 100 m

34. A 0.5 kg ball moving at 20 m/s is stopped in 0.1 s by a player. The force on the hand is: 1

(a) 10 N **(b)** 100 N **(c)** 50 N **(d)** 1000 N

35. In a lever the effort arm is 40 cm and the load arm is 10 cm. The effort needed to lift a 200 N load is: 1

(a) 800 N **(b)** 200 N **(c)** 50 N **(d)** 20 N

36. In a sound wave, the distance between a compression and the next rarefaction is: 1

(a) two wavelengths **(b)** one wavelength
(c) zero **(d)** half a wavelength

37. Which organelle is absent in animal cells? 1

(a) chloroplast **(b)** ribosome **(c)** mitochondrion **(d)** Golgi

38. The structure that reduces friction at joints is: 1

(a) bone **(b)** tendon
(c) muscle **(d)** cartilage and synovial fluid

39. Two species placed in the same genus are: 1

(a) unrelated
(b) more closely related than species in the same family but different genera
(c) of different kingdoms
(d) identical

40. A cell of the human body has 46 chromosomes. After mitosis each daughter cell has: 1

(a) 46 **(b)** 92 **(c)** 23 **(d)** 44

41. The forest absorbs CO2 at 20 tonnes per hectare per year. For 50 hectares per year it absorbs: 1

(a) 400 tonnes **(b)** 70 tonnes **(c)** 2.5 tonnes **(d)** 1000 tonnes

42. The 'ideal gas' model works best at: 1

(a) high pressure and high temperature **(b)** low pressure and high temperature
(c) any condition **(d)** high pressure and low temperature

43. To separate a mixture of kerosene and water we use: 1

(a) magnet **(b)** separating funnel
(c) evaporation **(d)** distillation

44. The formula of calcium hydroxide is: 1

(a) Ca_2OH **(b)** $CaOH$ **(c)** $Ca(OH)_2$ **(d)** CaO_2H

45. Which element has a valency of 3? 1

(a) sodium **(b)** chlorine **(c)** aluminium **(d)** magnesium

46. A graph of velocity against time is a straight line passing through the origin. The body has: 1

(a) zero acceleration **(b)** uniform acceleration from rest

(c) uniform velocity **(d)** retardation

47. A block is pulled with a force of 15 N and moves with constant velocity. The friction on it is: 1

(a) 7.5 N **(b)** 30 N **(c)** 0 **(d)** 15 N

48. A ball dropped from 5 m rebounds to 3.2 m. The fraction of energy lost is: 1

(a) 64 per cent **(b)** 3.2 per cent **(c)** 36 per cent **(d)** 50 per cent

49. The loudness of sound is measured in: 1

(a) metres **(b)** hertz **(c)** watts **(d)** decibels

50. The cell division that increases the number of cells for growth is: 1

(a) fertilisation **(b)** meiosis **(c)** osmosis **(d)** mitosis

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND